

An alpha-defensin gene single nucleotide polymorphism modulates the gut microbiota and may alter the risk of acute graft-versus-host disease

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Summary

We previously reported a protective association between single nucleotide polymorphisms (SNPs; rs4415345G and rs4610776A alleles) of Paneth cell α -defensin-5 against acute graft-versus-host disease (aGVHD). Because dysbiosis has been associated with aGVHD, we hypothesized that these SNPs may have a gut microbiota signature. In Lasso regression analysis of 248 healthy individuals, rs4415345G was associated with a higher abundance of *Odoribacter splanchnicus*, an anaerobic butyrogenic commensal. In multivariable analysis of data from 613 allogeneic haematopoietic cell transplant recipients, peri-engraftment presence of *O. splanchnicus* was associated with ~50% lower risk for grade II–IV aGVHD (hazard ratio 0.53, 95% confidence interval 0.28–1.00, $P = 0.05$). *O. splanchnicus* may protect rs4415345G individuals against aGVHD.

Keywords: graft-versus-host disease, paneth cell, microbiota, *odoribacter*, polymorphism.

Via the secretion of antimicrobial peptides (AMPs) such as defensins and regenerating islet-derived protein-3 α , small intestinal Paneth cells regulate the gut microbiota (Salzman *et al.*, 2009). α -Defensin-5 (HD5) accounts for 70% of the bactericidal peptide activity of Paneth cells (Ayabe *et al.*, 2000). Interactions between the host and microbiota at the gut mucosal barrier regulate mucosal and systemic immune responses and, in pathological states, may contribute to disease states such as acute graft-versus-host disease (aGVHD) after allogeneic haematopoietic cell transplantation (allo-HCT) (Staffas *et al.*, 2017). The hallmark of acute GVHD is alloimmune inflammatory damage to host

epithelial cells, with the gut being a primal target tissue (Zeiser & Blazar, 2017).

We previously reported an association between two single nucleotide polymorphisms (SNPs rs4415345 [G allele] and rs4610776 [A allele]) in the gene for HD5 (*DEFA5*) and lower incidence of aGVHD (Rashidi *et al.*, 2018). Although the functional consequences of these SNPs have not been studied, we suspected that these SNPs may influence HD5 protein levels. Therefore, considering that Paneth cell function is partly genetically regulated (Gulati *et al.*, 2012) and faecal concentrations of α -defensins are surrogate markers for gut microbial homeostasis (Eriguchi *et al.*, 2015), we

hypothesized that rs4415345 and rs4610776 may be associated with specific gut microbiota signatures.

First, we used a linked host-microbiota genetic database to test our hypothesis in healthy individuals. Metagenomic data were generated by the Human Microbiome Project (HMP), which comprises samples from 300 healthy adult volunteers (age 18–40) recruited from two geographic locations in the US. We downloaded the available operational taxonomic unit (OTU) table comprising normalized relative abundance data from <https://www.hmpdacc.org/hmsmcp2/>. Taxonomic profiling was conducted with MetaPhlAn2 (<http://huttenhower.sph.harvard.edu/metaphlan/>) and strain-level characterization was performed with StrainPhlAn (<http://huttenhower.sph.harvard.edu/metaphlan/>). Because some individuals had multiple Sequence Read Archive (SRA) IDs, we averaged the OTU abundances across IDs. All data were submitted to the Data Repository for the University of Minnesota at <https://doi.org/10.13020/9wn7-9142>. Subject genetic variation was provided as part of dbGaP project #19406. We subsetted the available HMP subjects to those with samples in the OTU table and focussed on our rs4415345 and rs4610776. We coded each genotype according to the presence of the allele in question (G for rs4415345 and A for rs4610776). Therefore, rs4415345 genotypes were coded as 0 (AA) vs. 1 (GG or GC), and rs4610776 genotypes were coded as 0 (TT) vs. 1 (AA vs AT). After matching metagenome samples to the genotyped individuals, we retained 248 subjects for the association study. Major allele frequencies at rs4610776 and rs4415345 were 0.73 and 0.67, consistent with the 1000 Genomes Project.

With the assumption that many microbial taxa have no association with a given SNP, we used least absolute shrinkage and selection operator (Lasso) regression as a test of association between host genotype and taxon abundance. This method permits variable selection by ‘shrinking away’ taxa that have no association. We used binary logistic regression to evaluate the association between taxonomic abundances (independent variable; arcsine square-root transformed) and each coded genotype (dependent variable). We implemented Lasso using the *glmnet* package (R Foundation for Statistical Computing, Vienna, Austria), with 10-fold cross-validation across multiple values of the tuning parameter lambda, using area under the curve (AUC) as the measure of error. We identified the value of lambda associated with the maximum AUC value (AUC_m), that is, the value yielding the best prediction. Because Lasso is sensitive to the training data used, we re-ran the regression with cross-validation 1000 times, each time retaining the AUC_m value. To build a null distribution for AUC_m , we again performed 1000 Lasso regressions with cross-validation, permuting genotype encodings before each test. Our significance test determined whether the mean AUC_m from the 1000 Lasso regressions on our true data fell outside the AUC_m distribution from permuted runs. As indicated in Table SI, although the mean AUC_m of unpermuted data was within 95% of the

permuted AUC_m distribution for both loci, the *P* value of 0.066 for rs4415345 was suggestive of an association. Therefore, we focussed our additional analyses on this locus.

To test whether specific taxa were repeatedly retained as predictors for rs4415345G (repeatability analysis), we recorded the taxa list with the highest AUC_m for each simulation. One thousand simulations were performed. We repeated the same analysis for permuted samples. If the regression performs better than random taxa selection, we expect the selected taxa to occur more frequently in the unpermuted dataset than in the permuted dataset. We observed very high repeatability; the frequency of selection of taxa as predictors fell almost entirely outside the distribution of the null distribution. Specifically, a single taxon (*Odoribacter splanchnicus*) was present in 99% of the simulations. Next, we performed stability selection analysis using R packages stabs and lars (cut-off 0.6, per family error rate 2) to determine how frequently a taxon is predictive of rs4415345G in Lasso across 100 random subsamples of the data. *O. splanchnicus* was again the only taxon that exceeded the recommended frequency cut-off of 0.6 (Fig 1A), with a positive association. Therefore, a higher relative abundance of this species was associated with the G allele of rs4415345. Finally, because Lasso selects only one of multi-collinear predictors, we tested whether the relative abundance of *O. splanchnicus* correlated strongly with any other species. The Spearman’s correlation coefficient was consistently <0.50, confirming the absence of additional taxa in association with rs4415345.

So far, we have shown an association between rs4415345 G allele and less aGVHD in HCT recipients (Rashidi *et al.*, 2018) and an association between the same allele and higher *O. splanchnicus* abundance in healthy individuals (current work), suggesting that this taxon may mediate the observed association in HCT recipients (protective effect). Next, we evaluated whether *O. splanchnicus* presence in the stool of HCT patients is associated with less aGVHD. The patients were admitted to the Memorial Sloan Kettering Cancer Center from 2009 to 2018 and enrolled in a biorepository protocol approved by the institutional review board (Table SII). For the analysis of baseline samples (earliest sample collected per patient between days –30 and –6), the landmark was HCT day 0 (when cells were infused). For the analysis of peri-engraftment samples (median abundance in all samples collected per patient during days 7–21), the landmark was HCT day 21. The genomic 16S ribosomal RNA V4/V5 variable region was amplified and sequenced on the Illumina MiSeq (San Diego, CA, USA) platform. Quality-filtered sequences with >97% identity were grouped into OTUs. Among 226 patients with evaluable baseline samples, *O. splanchnicus* was present in 65 and absent in 161 patients. Among 432 patients with evaluable peri-engraftment samples who had not been diagnosed with grade II–IV aGVHD as of the landmark, *O. splanchnicus* was present in 31 and absent in 401 patients. Among the 450 patients with evaluable peri-

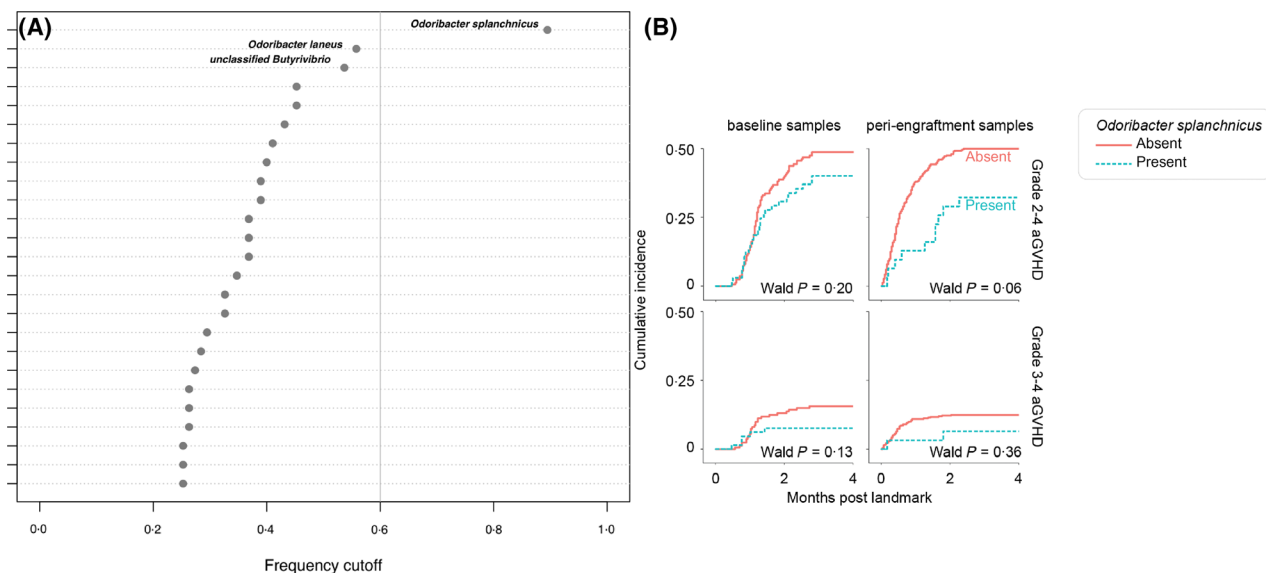


Fig 1. Association between rs4415345, gut microbiota, and acute graft-versus-host disease (aGVHD). (A) Stability selection analyses for the G allele of rs4415345. We chose the lower bound (0.6, represented by the vertical grey line) of the frequency cut-off recommended by the authors of the stabs R package. Each taxon is represented by a circle. Numbers along the x-axis show the frequency at which taxa were predictive in simulations across the 100 subsamples. The names of the top three taxa are shown. (B) Presence of *Odoribacter splanchnicus* in baseline and peri-engraftment samples. The horizontal axis indicates months post the landmark. For the analysis of baseline samples (earliest sample collected per patient between day -30 and -6), the landmark was haematopoietic cell transplantation (HCT) day 0 (when cells were infused). Among 226 patients with evaluable baseline samples, *O. splanchnicus* was present in 65 and absent in 161 patients. For the analysis of peri-engraftment samples (median abundance in all samples collected per patient during days 7–21), the landmark was HCT day 21. Among 432 patients with evaluable peri-engraftment samples who had not been diagnosed with grade II–IV aGVHD as of the landmark, *O. splanchnicus* was present in 31 and absent in 401 patients. Among the 450 patients with evaluable peri-engraftment samples who had not been diagnosed with grade III–IV aGVHD as of the landmark, *O. splanchnicus* was present in 31 and absent in 419 patients. [Colour figure can be viewed at wileyonlinelibrary.com]

engraftment samples who had not been diagnosed with grade III–IV aGVHD as of the landmark, *O. splanchnicus* was present in 31 and absent in 419 patients.

The results of univariate cumulative incidence analysis with grade II–IV or III–IV aGVHD as the dependent variable and baseline or peri-engraftment presence of *O. splanchnicus* as the independent variable are shown in Fig 1B. Relapse and non-GVHD death were considered as competing risks. The strongest association in univariate analysis was between peri-engraftment presence of *O. splanchnicus* and grade II–IV aGVHD (hazard ratio [HR] 0.54, 95% confidence interval [95% CI] 0.29–1.02, $P = 0.06$). A predetermined set of covariates was included in multivariable regression (Table I). In multivariable analysis, peri-engraftment presence of *O. splanchnicus* was associated with ~50% lower risk for grade II–IV aGVHD (HR 0.53, 95%CI 0.28–1.00, $P = 0.05$). Although the P values for *O. splanchnicus* in all other models were >0.05 , the results were consistently in the same direction, with *O. splanchnicus* presence showing a protective association (Fig 1B; Table I).

In summary, we found that the G allele of rs4415345, a SNP in the gene for Paneth cell α -defensin-5, is associated with the higher relative abundance of *O. splanchnicus* in healthy individuals and less aGVHD in HCT recipients. In

an independent dataset, we found that the peri-engraftment presence of *O. splanchnicus* was associated with a 50% reduction in risk for grade II–IV aGVHD. *O. splanchnicus* presence in baseline samples was less powerfully predictive of GVHD risk. The peri-engraftment microbiota may be a stronger predictor of immunologic outcomes because the interaction between microbiota and the graft may be maximal in the interval surrounding engraftment. *O. splanchnicus* is an anaerobic bacterium in the phylum *Bacteroidetes* (Hardham *et al.*, 2008) and present at low abundance in the gut of most individuals. *O. splanchnicus* has beneficial effects on the host, partly related to its butyrogenic activity (Werner *et al.*, 1975). Butyrate is a short-chain fatty acid implicated in protection of the gut barrier and reduction of aGVHD (Toubai *et al.*, 2016). In addition, *O. splanchnicus* produces indole from tryptophan (Göker *et al.*, 2011); indole produced by the gut microbiota is known to limit GVHD (Swimm *et al.*, 2018). *O. splanchnicus* presence is also correlated with the host's lower inflammatory cytokine production capacity (Schirmer *et al.*, 2016). *O. splanchnicus* is depleted in patients with inflammatory bowel disease (Morgan *et al.*, 2012). Based on the results of the present study and our previous report (Rashidi *et al.*, 2018), we propose that a higher abundance of *O. splanchnicus* in patients with the G allele of

Table I. Multivariable analysis of the association between *Odoribacter splanchnicus* and acute graft-versus-host disease (GVHD).

	HR (95%CI) Grade II–IV aGVHD	P	HR (95%CI) Grade III–IV aGVHD	P
Baseline <i>O. splanchnicus</i>				
Absent	Reference		Reference	
Present	0.76 (0.48–1.19)	0.23	0.46 (0.17–1.21)	0.12
Graft source				
Unmodified bone marrow	Reference		Reference	
Unmodified peripheral blood	1.14 (0.47–2.75)	0.77	0.82 (0.17–3.99)	0.81
Cord blood	1.38 (0.15–12.5)	0.78	0.28 (0.02–4.23)	0.36
Conditioning intensity				
Myeloablative	Reference		Reference	
Reduced intensity	1.26 (0.72–2.22)	0.42	1.34 (0.49–3.67)	0.57
Non-myeloablative	0.57 (0.22–1.11)	0.10	0.28 (0.06–1.38)	0.12
HCT-CI (continuous variable)	0.94 (0.86–1.04)	0.26	0.99 (0.83–1.19)	0.94
GVHD prophylaxis				
CNI + MMF-based	Reference		Reference	
CNI + MTX-based	0.58 (0.08–4.26)	0.59	0.27 (0.03–2.11)	0.21
PT-Cy-based	0.83 (0.10–7.00)	0.86	0.24 (0.02–2.97)	0.27
Peri-engraftment <i>O. splanchnicus</i>				
Absent	Reference		Reference	
Present	0.53 (0.28–1.00)	0.05	0.50 (0.12–2.06)	0.34
Graft source				
Unmodified bone marrow	Reference		Reference	
Unmodified peripheral blood	1.19 (0.73–1.94)	0.49	1.11 (0.43–2.88)	0.83
Cord blood	1.70 (0.22–13.16)	0.61	0.56 (0.06–5.28)	0.61
Conditioning intensity				
Myeloablative	Reference		Reference	
Reduced intensity	0.93 (0.64–1.35)	0.71	1.08 (0.52–2.22)	0.84
Non-myeloablative	0.38 (0.22–0.64)	<0.01	0.40 (0.13–1.26)	0.12
HCT-CI (continuous variable)	0.97 (0.90–1.03)	0.30	1.04 (0.91–1.18)	0.57
GVHD prophylaxis				
CNI + MMF-based	Reference		Reference	
CNI + MTX-based	0.72 (0.10–5.22)	0.75	0.37 (0.05–2.74)	0.33
PT-Cy-based	1.00 (0.13–7.70)	1.00	0.25 (0.02–2.66)	0.25

aGVHD: acute graft-versus-host disease; CI: confidence interval; CNI: calcineurin inhibitor; HCT-CI: haematopoietic cell transplantation-specific comorbidity index; HR: hazard ratio; MMF: mycophenolate mofetil; MTX: methotrexate; PT-Cy: post-transplant cyclophosphamide.

rs4415345 is protective against colitis and aGVHD. Host genetics-mediated regulation of the abundance of *O. splanchnicus* was suggested in a previous twin study (Lim *et al.*, 2017), albeit with unknown mechanisms. Our findings suggest Paneth cell's most abundant AMP as a key mediator, consistent with physiologic modulation of the gut microbiome by Paneth cell defensins (Ehmann *et al.*,). Future research should evaluate the effect of the SNP on HD5 gene expression, the effect of HD5 on *O. splanchnicus*, and the association between rs4415345 and *O. splanchnicus* in HCT patients.

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AR designed the study and wrote the manuscript. ALCG, JUP, DGB and RRJ collected and analyzed data from the clinical cohort. AH analyzed HMP data. CS, BRB and DJW

critically reviewed the results and enhanced the manuscript. We thank Joshua Baller for help with dbGaP setup and methodological advice, as well as Evan Bollig for assistance with the Stratus computing environment required for protected data.

Competing interests

The authors have no competing interests.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table SI. Permuted and unpermuted AUC_m distributions.

Table SII. Patient characteristics in the HCT cohort.